

The background features a dark blue gradient with several vertical orange bars of varying heights. Overlaid on these are white lines with circular markers at various points, creating a network-like or data-path appearance. Some of these lines and markers are slightly blurred, giving a sense of depth and motion.

SHOPSPHERE – RETAIL DATA ANALYTICS

Retail Sales Analytics Using
Excel, SQL, Power BI and
Python

Problem Statement:

In a competitive retail market, businesses need to track sales performance.

understand customer behavior, and identify growth opportunities.

This project aims to build an end-to-end analytics solution that enables insights generation from raw transactional data using a multi-tool approach.

Main Objective of Analysis



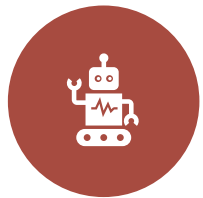
Data Cleaning and
Preprocessing using
Excel



Data Storage, Querying,
and Transformation using
SQL



Data Analysis and
Visualization using Python
(Pandas, NumPy,
Matplotlib/Seaborn)



Customer Segmentation
using Machine Learning



Business Intelligence and
Reporting using Power BI



End-to-End Data Pipeline
Integration

Excel – Data Cleaning & Preparation

Standardize Column Headers(e.g., use snake_case or camelCase).

Remove Duplicates Identify and remove any duplicate records (e.g., in customers order_items).

Handle Missing Values Use filtering and formulas to find blanks/nulls.

Fill missing ZIP codes or phone numbers with placeholders or “NA”.

Add a column in order_items to calculate total_price = (list_price * quantity) - discount.

Create Basic Pivot Table : Build a pivot table summarizing total sales by product category.

VLOOKUP and XLOOKUP are used to merge data from different tables by matching a common ID.

SQL – Database Management and Querying



Create Tables Based on ERD



Use CREATE TABLE statements to replicate the exact structure of the ER diagram (with constraints).



I used SQL queries with JOIN, GROUP BY, ORDER BY and aggregation functions to analyze the data.



Top 5 Selling Products Use ORDER BY and LIMIT to get the top 5 most sold products by quantity.



I also created a customer segmentation table called customer_segments, which was later used for the Python machine learning result

Python and Pandas

Load Data from SQL

Use pandas . read sql() to pull the orders, order items, and customers tables into a Data Frame.

Basic EDA (Exploratory Data Analysis) Use df .describe(), df.info(), and df. Value counts() to summarize the dataset.

Plot basic charts using matplotlib or seaborn.

Calculate RFM Features for Customers Compute Recency, Frequency, and Monetary values for each customer.

Recency: Days since last order

Frequency: Number of orders

Monetary: Total value of all purchases

Export Segmentation Results to SQL

Power BI – Visualization & Dashboarding

- I used Power BI to create an interactive dashboard for my Retail Sales Analytics.
- I connected and transformed the data, created DAX measures, and used different visualizations such as charts, KPI cards, tables, and slicers.
- The dashboard helps analyze sales trends, product performance, customer behavior, store performance, and staff revenue.
- Finally, Power BI helped me convert raw data into meaningful business insights.

Results:

- Fully cleaned, relational dataset stored in SQL
- Insightful analytics using Python
- Segmentation-based insights using ML
- Interactive dashboard in Power BI
- Complete ETL pipeline from Excel to Power BI.

